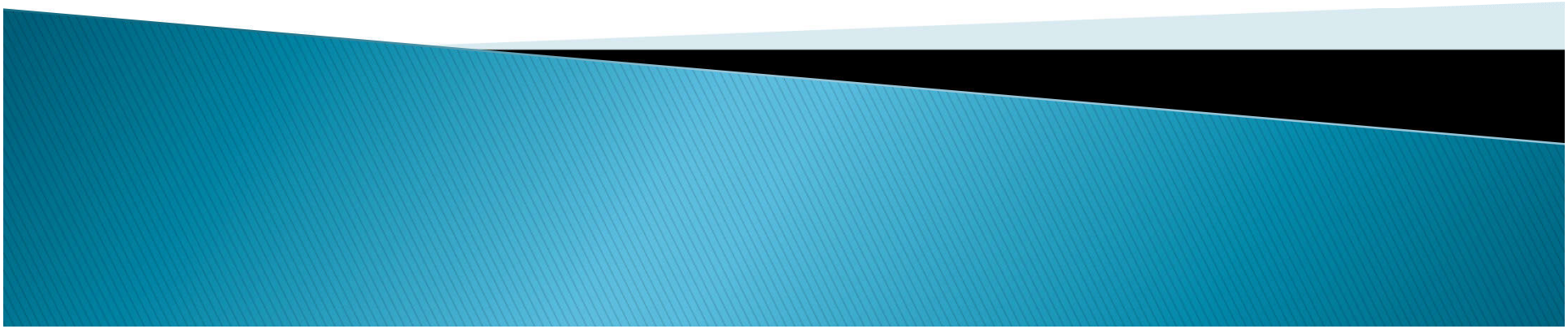
Data Mining in a nutshell

Nazerfard, Ehsan
nazerfard@aut.ac.ir



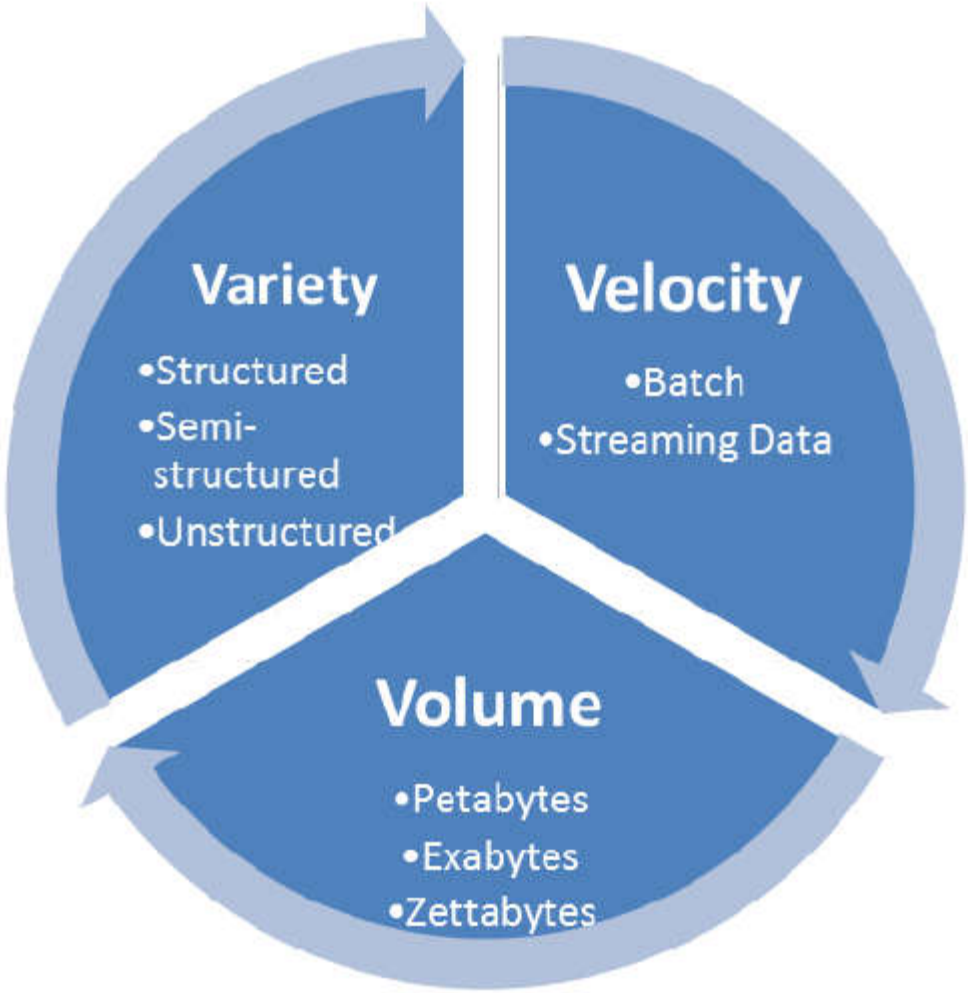
Data Mining

- **Data mining** is an area that has taken much of its inspiration and techniques from machine learning (and some, also, from statistics), but is put to different *ends*.

  Data mining is carried out by a *person*, in a specific situation, on a particular data set, with a goal in mind.

  *Note: The goal is generally *not* to develop a more sophisticated understanding of the underlying data generating process. Quite often, the data set is *massive.*

Big Data


Variety
•Structured
•Semi-structured
•Unstructured
Velocity
•Batch
•Streaming Data
Volume
•Petabytes
•Exabytes
•Zettabytes

What is Data Mining?



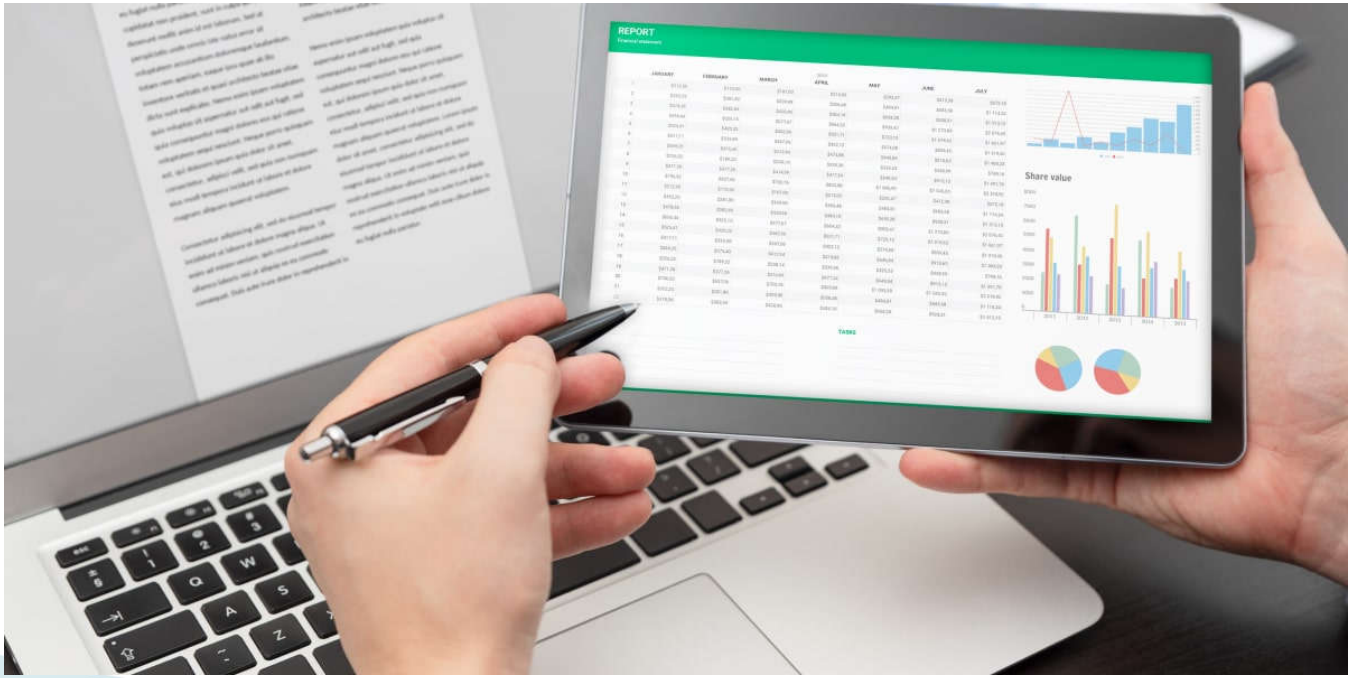
** word cloud (a visualization technique)

What is Data Mining?



Data Mining Tools

- Python
- R
- RapidMiner
- Orange
- Knime
- …



The Iris Dataset

archive.ics.uci.edu/dataset/53/iris



Iris

Donated on 6/30/1988

A small classic dataset from Fisher, 1936. One of the earliest known datasets used for evaluating classification methods.

Dataset Characteristics	Subject Area	Associated Tasks
Tabular	Biology	Classification

Feature Type	# Instances	# Features
Real	150	4

Dataset Information

What do the instances in this dataset represent?

Each instance is a plant

Additional Information

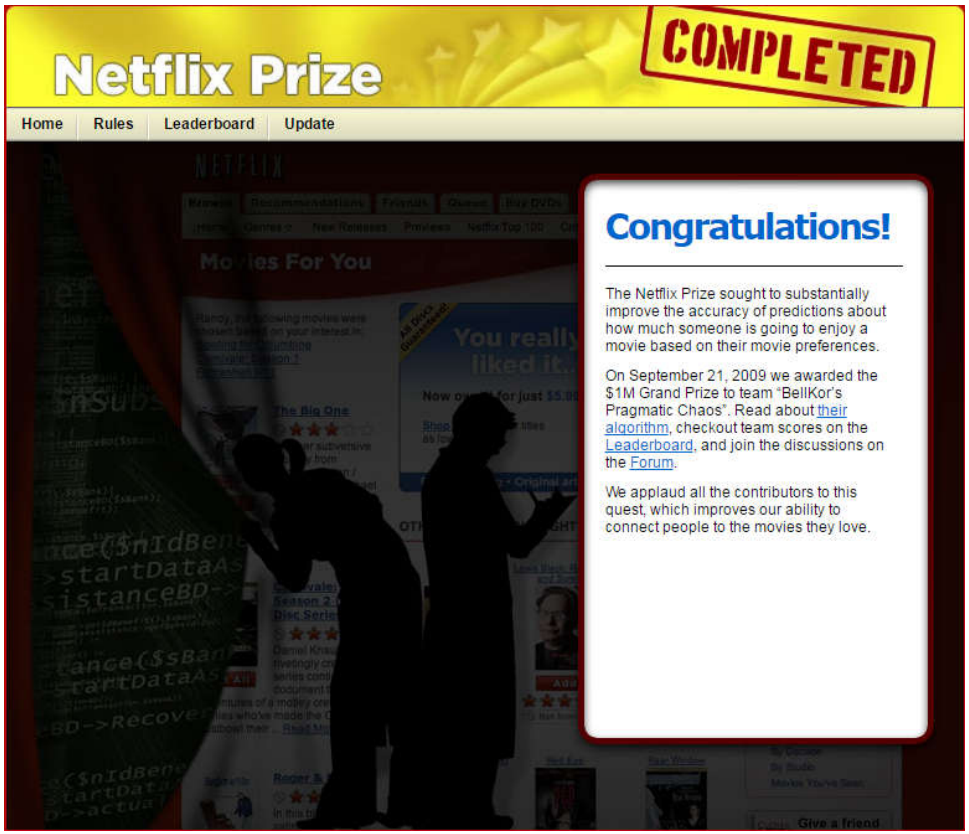
This is one of the earliest datasets used in the literature on classification methods and widely used in statistics and machine learning. The data set contains 3 classes of 50 instances each, where each class refers to a type of iris plant. One class is linearly separable from the other 2; the latter are not linearly separable from each other....

SHOW MORE

Has Missing Values?

No

Netflix Prize



Matrix completion

(collaborative filtering)

DM Scholars



Trevor Hastie
Professor of Statistics, Stanford University
Verified email at stanford.edu
Cited by 148166
Statistical learning and modeling data mining machine learning



Jiawei Han
Abel Bliss Professor of Computer Science, University of Illinois
Verified email at cs.uiuc.edu
Cited by 133869
data mining database systems data warehousing information networks



Jeffrey Ullman
Professor of Computer Science, Stanford University
Verified email at stanford.edu
Cited by 114190
Database Systems Data Mining



Rakesh Agrawal
Technical Fellow, Microsoft Research
Verified email at microsoft.com
Cited by 106322
Data Mining Web Search Education Privacy



Kim Minwoo
University of Central Arkansas
Cited by 85381
AI Data Mining Algorithm Bioinformatics

DM Scholars (cont.)



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Professor of Computer Science, University of Illinons at Chicago
Verified email at cs.uic.edu
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Data mining Database Privacy



John Canny
University of California, Berkeley
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Cited by 77853
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Vipin Kumar
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Verified email at cs.umn.edu
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Data mining parallel computing bioinformatics climate informatics graph partitioning



Chih-Jen Lin
Professor of Computer Science, National Taiwan University
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Cited by 70176
Machine learning Data Mining Optimization Artificial Intelligence

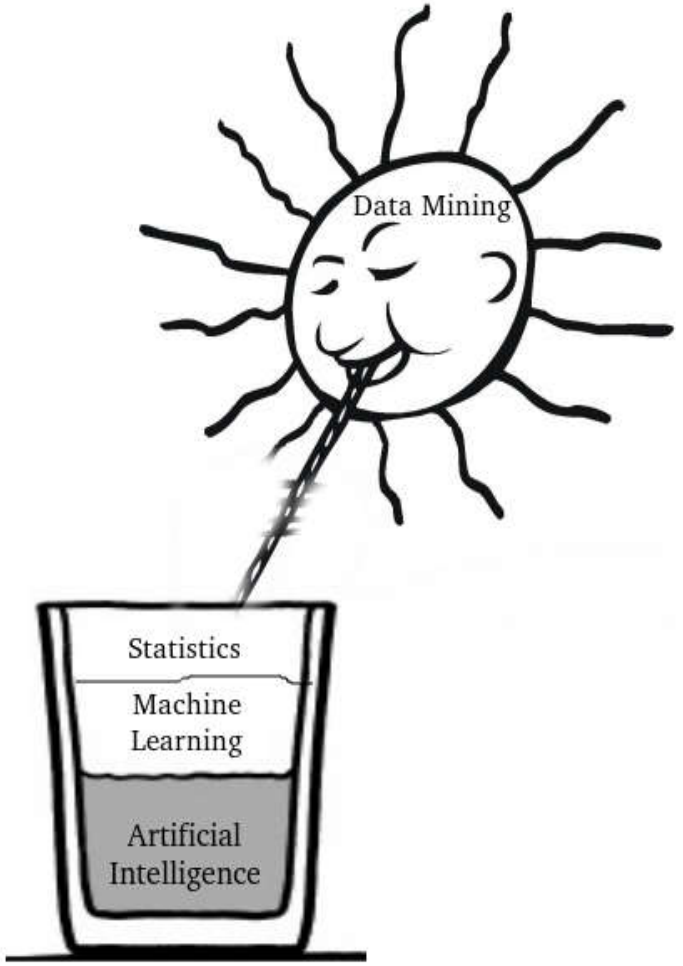


Jian Pei
Professor of Computing Science, Simon Fraser University
Verified email at cs.sfu.ca
Cited by 69456
Data mining big data analytics database systems information retrieval

Related Fields of Study

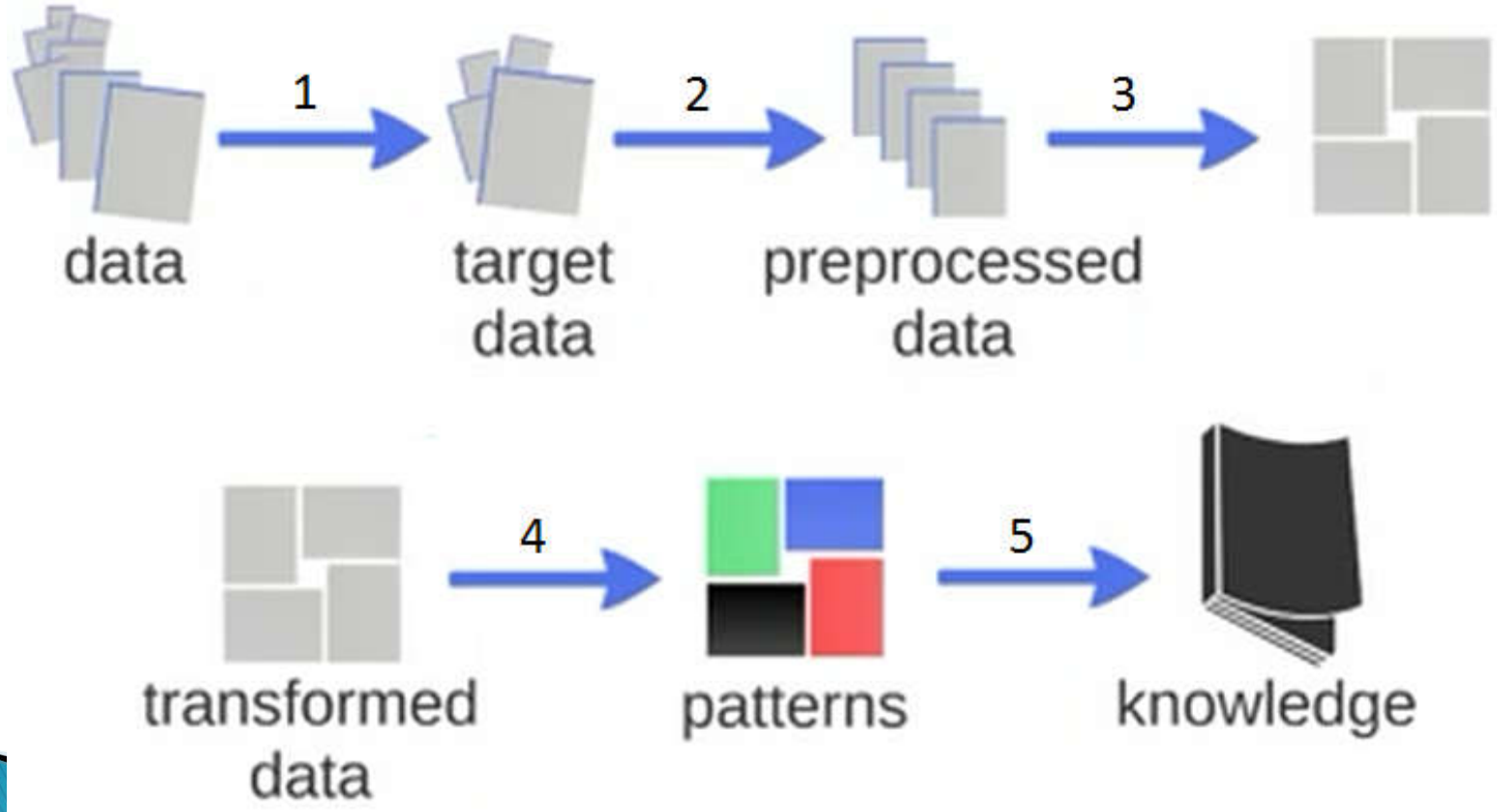
Making friends with AI

(highlighting attention)



Data Mining Steps

- مراحل عمومی روشهای داده‌کاوی



Visualization

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Play our Spelling Bee
Take a spelling challenge that is both vocabulary-expanding and downright addictive.

Link